

ネットワーク計画ツールとしての欲求経路創発シミュレーションマルチエージェントシステムの開発と評価

Development and Evaluation of a Multi-Agent System simulating Emergence of Desire Paths as Networks Planning Tool

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1 Introduction

Desire paths emerge from repeated movements of individuals through specific locations, where paths form through continual traversals. The multi-agent model in this research simulates this phenomenon by deploying numerous agents that interact with their environment to create a global network pattern. Previous models, such as those based on pedestrian movement [1] and slime mold approximations [2], have been instrumental in addressing transportation network problems. However, these models have limitations when applied to designing efficient human path systems.

The slime mold model mimics exploration and contraction processes but often results in random disconnections and regional faults within the network. Meanwhile, pedestrian movement models more accurately replicate human behavior but fail to account for key aspects such as pattern stability, path relevancy, and traffic density distribution. Additionally, the time factor is crucial in transportation problems, as human traffic demands constantly change and must be considered in network design [3]. The evolving nature of human traffic and its impact on network efficiency and structure remains underexplored in current models.

This research seeks to address these gaps by developing a comprehensive model that simulates desire path evolution while incorporating time variability, pattern stability, and network performance.

2 Goals

My research focuses on developing a multi-agent model capable of creating optimized solutions for diverse applications. This involves identifying critical parameters that influence agent behavior and overall performance, enabling a deeper understanding of how these factors shape outcomes. By evaluating the performance

of path systems based on various agent configurations, the research aims to uncover insights into system efficiency and adaptability. At term, the goal is to provide a practical and robust tool that can be used to design and optimize future systems tailored to a wide range of scenarios and applications.

3 Human visitation law

When agents reach their current target, they select a new biased random target based on their current location. This decision is influenced by a model inspired by human visitation patterns, which suggests that certain locations are more likely to be visited due to their attractiveness and frequency of past visits. The target selection process follows the human visitation law, which takes into account the distance of a potential target from the agent's current position, the frequency of visits of that location and the inherent attractiveness of the target itself.

The target selection is governed by the following equation:

$$\rho_i(r, f_i) = \frac{\mu_i}{(rf_i)^2} \quad (1)$$

In this equation, ρ_i represents the weight or attractiveness of hotspot i , which determines the likelihood of the agent selecting it as their next target. The variable r represents the distance between the agent's current location and the potential target, while f_i is the frequency with which the target has been visited by agents in the past. μ_i is the attractiveness factor of the hotspot, which could be based on factors such as its resources, importance, or any other characteristic that makes it appealing to agents.

This biased random selection process ensures that agents are more likely to choose targets that are closer, less frequently visited, or more attractive, while still al-

lowing for some randomness to ensure diversity in the emerging network patterns. Over time, this approach leads to the creation of efficient networks that reflect the combined influence of local agent behavior and global environmental factors.

4 Agent guiding principle and morphology

Each agent has a view section area determined by the vision depth (SO), the width (SW) and the view angle (SA), as shown in Figure (1). This vision area aims to analyze the environment, run a patch-finding procedure and determine the best moving direction according to weighted global patch direction and relative target direction. If paths are detected, agent evaluates patches that will allow him to get closer to its target, according to Equation (2). Otherwise, the agent adopt the strategy of the direct route towards its destination. During its movement, the agent deposits a patch value in the environment, which impacts the behavior of the other agents. The impact of an agent depends on the proportion of complete travels among the total travels achieved in the previous generation. The more an agent completes travels, the more impact it has on the next generation. At the end of each generation, all agents are relocated to their starting hotspot.

$$w_i = p_{aff}(i) \frac{1 + \vec{T} \cdot \vec{P}}{2} \quad (2)$$

where w_i is the path weight of the pixel i , $p_{aff}(i)$ is the path affordance at pixel i , $\vec{T}$ is the target direction and $\vec{P}$ is the pixel direction. Pixel and target direction are relative to agent position in the environment.

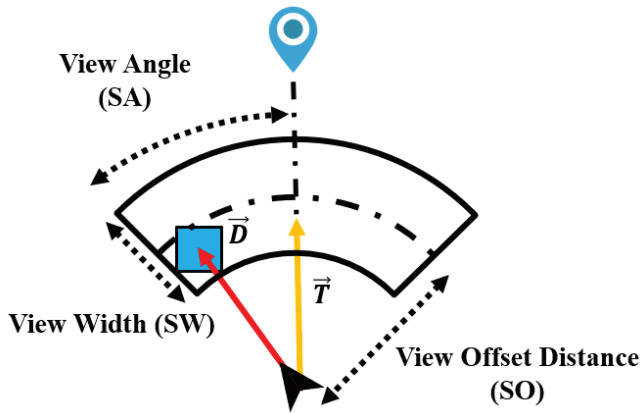


Figure 1: Agent morphology

When deploying thousands of agents in a given environment configuration, agents are following the highest afforded path, which result in putting in common paths and shrink the total paths system to a relative stable state.

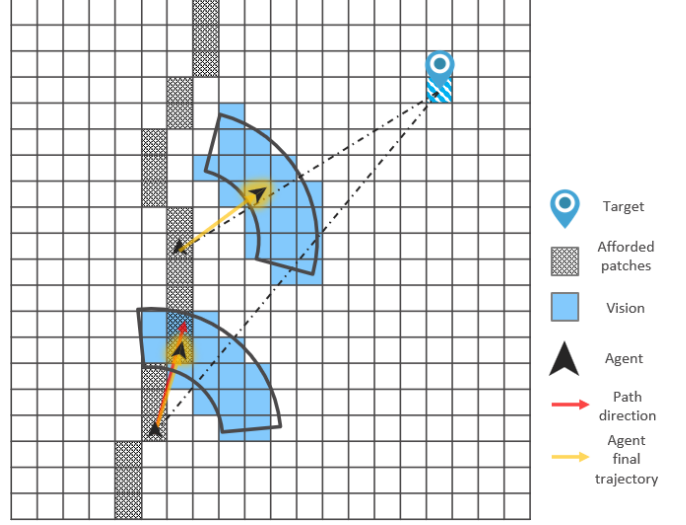


Figure 2: Guiding principles of agent movement and emergent desire paths

5 Results

To demonstrate the convergence of the network pattern and the performance of the resulting path system, the model was tested on a flat surface with no pre-existing paths, using a randomly generated set of hotspots. The simulation deployed 1000 agents across a 2D grid of 400x400 pixels, with each generation consisting of 5000 iterations.

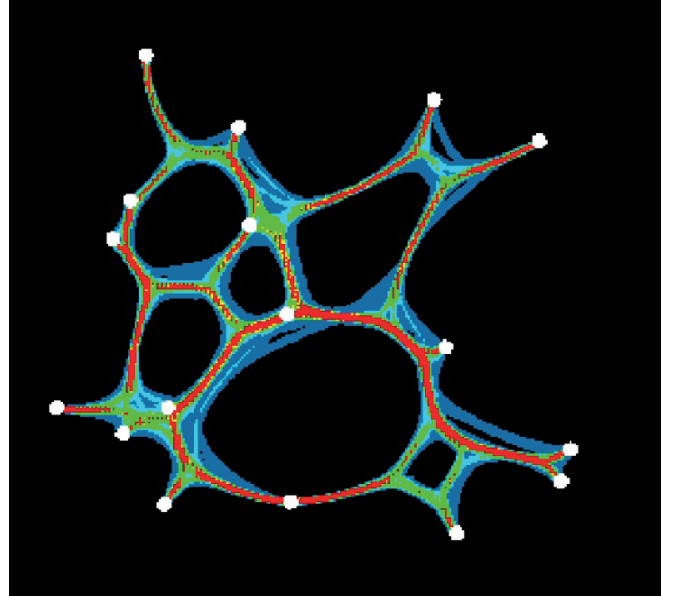


Figure 3: Resulting path system from simulation with a traffic density mask on created paths

Initially, agents follow straight routes to their targets. However, over time, their paths are influenced by the tracks left by other agents. As agents repeatedly traverse certain routes, they begin to favor the most fre-

quently used paths, gradually converging on areas that help them reach their targets more efficiently. This self-organizing behavior results in the emergence of highly traveled paths. As the simulation progresses, unused paths gradually decay, causing the network to shrink and focus on commonly shared routes, which are highlighted through a color scale.

The color scale represents the "affordance" of each path, reflecting both the traffic density of a route and its relevance for inclusion in the global network in the next generation. The affordance scale is defined as follows:

- Blue: Low traffic density, relevance score of 0.2
- Green: Moderate traffic density, relevance score of 0.4
- Yellow: High traffic density, relevance score of 0.6
- Red: Very high traffic density, relevance score of 1.0

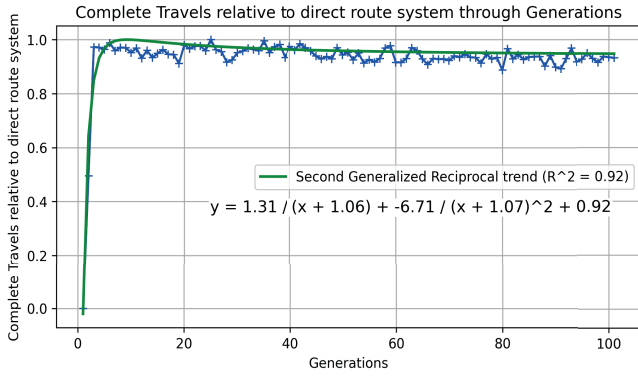


Figure 4: Evolution of complete travels at each generation

The total number of travels is directly influenced by the travel time of agents reaching their target. This travel time, in turn, is affected by several factors, including the duration of each generation in the simulation, the maximum speed of the agents, and the overall configuration of the environment. Notably, the offset distance between agents primarily impacts the scale of the resulting paths, as demonstrated in [2]. However, the view angle of the agents has a more significant effect, as it determines the resulting pattern's structure and orientation. Previous studies have shown that common desire paths typically emerge with angles ranging between 45° and 90°.

The convergence of the resulting path network is influenced by several key factors: the decay rate of deposits left by agents, the total number of agents, and the pseudo-random generation process used to select new targets. Specifically, the pseudo-random number is calculated based on the agent's unique ID and the simulation time, ensuring variability in target selection. Since existing paths influence the movement and trajectory of

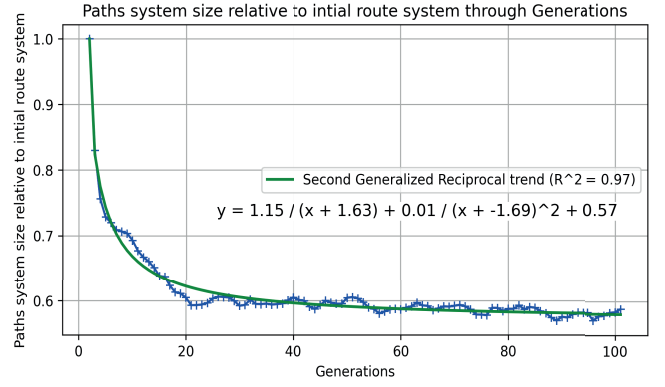


Figure 5: Paths system cost at each generation

agents, they also indirectly affect their arrival time at targets and the selection of subsequent targets.

Interestingly, the convergence of the pattern is less affected by the total number of completed travels across generations. Furthermore, after a certain threshold, the number of agents deployed has minimal impact on the density of the resulting routes. Deploying a substantial number of agents amplifies their influence on each other's tracks, which in turn accelerates the convergence of the path network.

In summary, while the behavior of individual agents and their interaction with the environment significantly shape the emergent desire path patterns, the system exhibits resilience to variations, focusing instead on the interplay between environmental factors and movement dynamics.

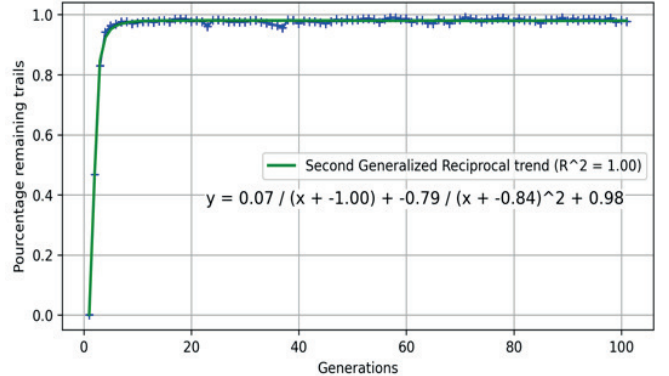


Figure 6: Network pattern convergence through generation

Another limitation of the convergence process lies in the distribution of afforded paths within the resulting pattern. Paths with higher affordance—those that are more frequently used or more accessible—are more likely to persist across successive generations. While a higher proportion of highly afforded paths contributes to a more stable pattern over time, it also slows down the convergence process, requiring more generations for the system to reduce to an optimal network configuration.

This slow convergence can sometimes lead to deadlocks in the pattern formation, where certain paths stabilize prematurely, preventing further optimization of the network. As a result, the system may become trapped in a local optimum, where the network structure is stable but not necessarily the most efficient or optimal solution.

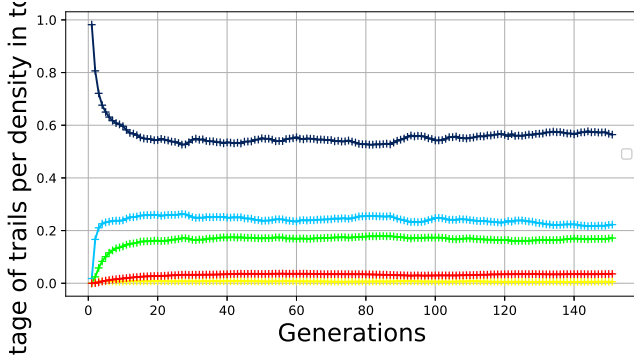


Figure 7: Proportion of trails per density in trail network through generations

Figure (7) illustrates the proportion of afforded paths in the total path system across generations. It shows that the majority of paths are lowly afforded and are more transient in nature, with their persistence heavily dependent on the decay rate of path deposits. When the decay rate is higher, these low-affordance paths tend to disappear more quickly, reducing their influence on the overall convergence of the path system. However, if the decay rate is too low, these paths may linger unnecessarily, delaying the system’s ability to converge to an optimal configuration.

Figure (8) shows the impact on path system performance according to different configuration of agent morphology. This highlights that the depth view doesn’t impact significantly the number of travels the system can achieve. Then, it points out that the scale of the simulation doesn’t impact the number of travels in the path system. However, the angle has more impact since it influences the trajectory of the agent and then, the shape of roads. Since a large number of agents are deployed, it has even more impact on the total size of the system.

6 Conclusion

This research successfully developed a multi-agent model capable of simulating optimized network designs by replicating the emergence of desire paths and the diffuse effect from slime mold. The evaluation demonstrated that while the depth view parameter does not significantly impact the completion of travels, it effectively adjusts the scale of the simulation, offering valuable flexibility in scenario modeling. These findings provide a solid foundation for advancing the study of agent-based network optimization.

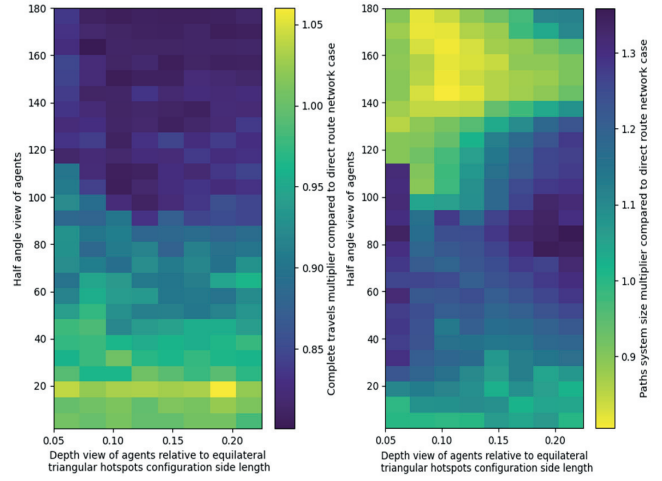


Figure 8: Path system performances according to different agent configurations. Complete travels (left), paths system size (right)

Future works will focus on optimizing the number of agents deployed within the simulation to enhance efficiency further. A critical next step will involve integrating real-world data, such as the Tokyo subway system, to compare simulation outcomes with actual railway network maps, deepening the model’s relevance and applicability to real-world transportation systems.

References

- [1] Simple agents complex emergent path systems agent-based modeling of pedestrian movement, Ma, L., Brandt, S A., Seipel, S., Ma, D., 2023
- [2] Jeff Jones, “A Morphological Adaptation Approach to Path Planning Inspired by Slime Mould” , Centre for Unconventional Computing, University of the West of England, Coldharbour Lane, Brisol, BS16 1QY, UK, 2015.
- [3] Physarum machines imitating a Roman road network the 3D approach, Vasilis Evangelidis, Jeff Jones, Nikolaos Dourvas, Michail-Antisthenis Tsompanas, Georgios Ch. Sirakoulis and Andrew Adamatzky, 2017
- [4] The universal visitation law of human mobility, Markus Schläpfer, Lei Dong, Kevin O’Keeffe, Paolo Santi, Michael Szell, Hadrien Salat, Samuel Anklesaria, Mohammad Vazifeh, Carlo Ratti and Geoffrey B. West, 2021
- [5] Rodrigo Fernandes Ferreira de Almeida, ” Formation and Optimisation of Vein Networks in Physarum” , 2023
- [6] Mitja Bezensek, Borut Robic , “A Survey of Parallel and Distributed Algorithms for the Steiner Tree Problem” International Journal of Parallel Programming volume 42, 2013.
- [7] Flow-network adaptation and behavior in slime molds, Audrey Dussutour, Chloe Arson, 2023
- [8] Slime Mold Inspired Distribution Network Initial Solution, Verner Püvi, Robert J. Millar, Eero Saarijärvi, Ken Hayami, Tahitoa Arbelot and Matti Lehtonen, 2020